ENGN 1580 - Communications Systems

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Chapter 1

Fourier Transforms

1.1 Jan 22

1.1.1 Linear Systems

What is a linear system?

1. $L[u+v] = L[u] + L[v]$ (additivity)
2. $L[au] = aL[u]$ (scaling)
3. $L[ax+by] = aL[x] + bL[y]$ (superposition)

Do we actually need scaling if we have additivity?

Example 1.

$$L[x+x] = Lx + Lx = 2Lx$$

seems like we don't. What about rationals?

Example 2. if we want $L[0.1x]$ we can see

$$L[x] = L[0.1x + \cdots + 0.1x] = 10L[0.1x] \implies L[0.1x] = \frac{1}{10}L[x]$$

still working...

Example 3. And even for negatives...

$$L[-x] = L[x-x-x] = L[x] + 2L[-x]0 = L[x] + L[-x]L[-x] = -L[x]$$

(**Remark:** this also immediately leads to the identity $L[0] = 0$ for all linear systems)

Example 4 (A counter): What if $L[ax]$ is irrational? We can approximate by rationals and apply the same arguments above but this requires L continuous on $\mathbb{R} \setminus \mathbb{Q}$ – a rather strong condition

Where things really break are $\mathbb{C}$: what could we possibly say about $L[jx]$? (convention: j is imaginary basis in this class)

1.1.2 Periodic Signals

Signal: a function

periodic signals: from fourier series

$$p(t) = \sum p_k exp(2\pi jk/T)$$

for period T where

$$p_k = \frac{1}{T}\int_{-T/2}^{T/2} p(t)e^{-j*2\pi/T*kt}\,dt$$

(as an aside notice that this is a projection)

Fourier transform:

$$\mathfrak{X}f = \int x(t)e^{-j*2\pi ft}\,dt$$

Inverse Fourier:

$$x(t) = \int \mathfrak{X}fe^{j*2\pi ft}\,df$$

where f is the frequency (note we are missing the coefficients because we work in Hz)

Laplace:

$$x(s) = \int x(t)e^{-st}\,dt$$

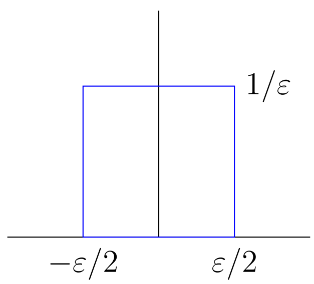
Notice that all of these transforms have integrals of exponentials. Why might that be?

Linear time invariant (LTI) systems: Let L[x(t)] = y(t). If this is an LTI,

$$L[x(t-\tau)] = y(t-\tau)$$

impulse: square function of lengths $1/\varepsilon$ centered at 0 satisfying

1. $\int \delta(t)\,dt = 1$
2. $\int \delta(t-a)s(t)\,dt = s(a)$(sifting property)



This leads to an alternative definition of an LTI $h(t)$ as one for which

1. $h(\delta(t)) = h(t)$
2. $h(a\delta(t-\tau)) = ah(t-\tau)$

Let us consider a general case

$$x(t) = \int x(\tau)\delta(t-\tau)d\tau$$

now what is $h(x(t))$?

If we view the integral as a Riemann sum, we can see easily that

$$\begin{aligned} h(x(t)) &= h(\int x(\tau)\delta(t-\tau)d\tau) \\ &= \int x(\tau)h(t-\tau)d\tau \\ &= y(t) \end{aligned}$$

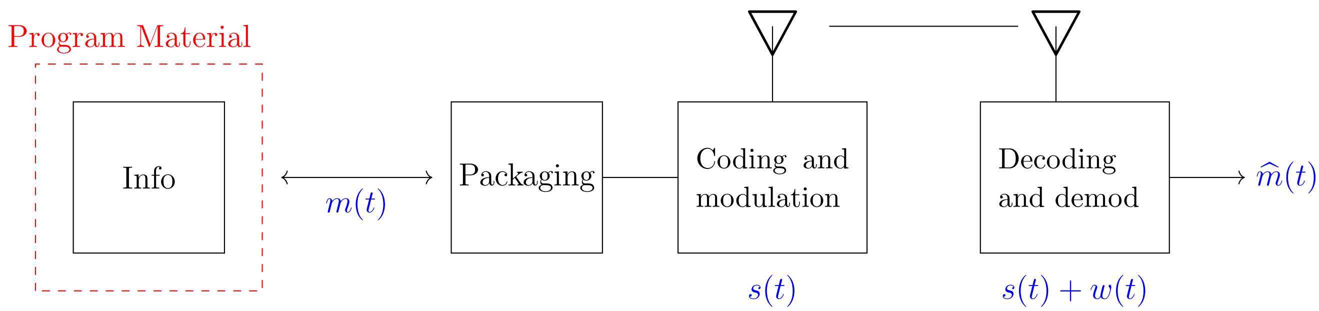
a convolution! (and of course we could equally see by a change of variables $\int x(t-\tau)h(\tau)d\tau$)

What about $h(e^{jst})$?

$$\int e^{js(t-\tau)}h(\tau)d\tau = e^{jst}\int h(t)e^{-jst}d\tau$$

1.2 Jan 27

Outline of a Communications System:



where:

- info is viewed broadly as songs, video, text, etc.
- $m(t)$ is the message signal (assuming no loss between program material and packaging)
- $s(t)$ is the transmitted signal
- $w(t)$ is noise added by the transmission interference
- $\widehat{m}(t)$ is the received (estimated) message signal

1.2.1 Linear system review:

A system L is **linear** if it satisfies *superposition*:

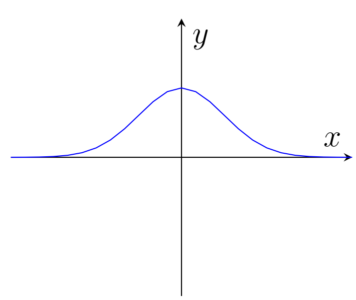
$$Lax + by = aL[x] + bL[y]$$

If we define a delta function $\delta(t)$ satisfying

1. $\int_{-\infty}^{\infty} \delta(t)\, dt = \int_{0^-}^{0^+} \delta(t)\, dt = 1$
2. $\int \delta(t-a)s(t)\, dt = s(a)$ (sifting property)

Last time we saw the square pulse. However, we can also define it as the limit of a Gaussian as $\sigma \to 0$:

$$\delta_\sigma(t) = \sqrt{\frac{1}{2\pi\sigma^2}}e^{-\frac{t^2}{2\sigma^2}}$$



Exercise: Show that $\delta_\sigma(t)$ is a delta function as $\sigma \to 0$.

Impulse response of a linear system:

$$\delta(t-t_0) \longrightarrow \boxed{h(t)} \longrightarrow h(t-t_0)$$

With a little work, we can generalize this property to see that for any input signal $x(t)$, the output of a linear system is given by the convolution of the input with the impulse response:

$$\begin{aligned}
y(t) &= h(x(t)) \\
&= \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\, d\tau \\
&= \int_{-\infty}^{\infty} x(t-\tau)h(\tau)\, d\tau
\end{aligned}$$

Fourier Transform:

$$h(t) \overset{\mathcal{F}}{\Longrightarrow} H(f) = \int_{-\infty}^{\infty} h(t)e^{-j\,2\pi f t}\, dt$$

Example: Let $x(t) = \cos 2\pi f_0 t$. What is $X(f)$?

$$\begin{aligned}
\cos 2\pi f_0 t &= \frac{e^{j\,2\pi f_0 t} + e^{-j\,2\pi f_0 t}}{2} \\
h(\cos 2\pi f_0 t) &= \frac{1}{2}h(e^{j\,2\pi f_0 t}) + \frac{1}{2}h(e^{-j\,2\pi f_0 t}) \\
&= \frac{1}{2}e^{j\,2\pi f_0 t}H(f_0) + \frac{1}{2}e^{-j\,2\pi f_0 t}H(-f_0)
\end{aligned}$$

Let's generalize:

$$\begin{aligned}
h(x(t)) = y(t) - \int x(\tau)h(t-\tau)\, d\tau \\
&\overset{\mathcal{F}}{=} \int\int x(\tau)h(t-\tau)e^{-j\,2\pi f t}\, dt\, d\tau \\
&= \int x(\tau)\left(\int h(t-\tau)e^{-j\,2\pi f t}\, dt\right) d\tau \\
&= \int x(t)\left[\int h(u)e^{-j\,2\pi f(u+\tau)}\right] d\tau \qquad (u = t-\tau) \\
&= \int x(t)\underbrace{\left[\int h(u)e^{-j\,2\pi f u}\right]}_{H(f)} e^{-j\,2\pi f \tau}\, d\tau \\
&\overset{\mathcal{F}^{-1}}{=} X(f)H(f)
\end{aligned}$$

Which yields the important result: *convolution in the time domain is multiplication in the frequency domain:*

$$x(t) \star h(t) = (x \star h)(t) \overset{\mathcal{F}}{=} X(f)H(f)$$

In fact, the work here leads to a number of useful properties of the Fourier transform:

Time shift property:

$$\boxed{\mathcal{F}[x(t-t_0)] = e^{j\,2\pi f t_0} X(f)}$$

Scaling:

$$\begin{aligned}
\mathcal{F}[x(at)] &= \int x(at) e^{-j\,2\pi f t}\,dt \\
&= \int x(u) e^{-j\,2\pi f (u/a)} \frac{du}{a} \qquad\qquad (du = a\,dt) \\
&= \boxed{\frac{1}{|a|} X(f/a)}
\end{aligned}$$

Differentiation:

$$\begin{aligned}
\frac{d}{dt} x(t) &= \frac{d}{dt}\left[\int X(f) e^{j\,2\pi f t}\,df\right] \\
&= \int X(f)(j\,2\pi f) e^{j\,2\pi f t}\,dt
\end{aligned}$$

so

$$\boxed{\mathcal{F}\left[\frac{dx}{dt}\right] = X(f)\; j2\pi f}$$

Integration:

$$\boxed{\mathcal{F}\left[\int_{-\infty}^{t} x(u)\,du\right] = \frac{X(f)}{j\,2\pi f} + \frac{X(0)}{2}\delta(f)}$$

Exercise: Prove the integration property

Parseval:

$$\boxed{\int_{-\infty}^{\infty} x^2(t) = \mathcal{E}_x = \int_{-\infty}^{\infty} |X(f)|^2\;df}$$

("the energy in time domain = energy in frequency domain")

We've seen over and over that $\mathcal{F}[x(t)] = X(f)$? What about $\mathcal{F}[X(f)]$?

First, two fact that should be obvious:

$$\begin{aligned}
\mathcal{F}[\delta(t)] &= 1 \\
\mathcal{F}[\delta(t-\tau)] &= e^{-j\,2\pi f t}
\end{aligned}$$

and since (up to sets of measure zero),

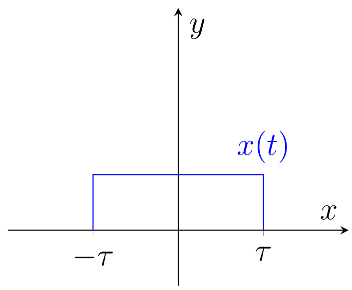
$$\delta(t-\tau) \underset{\mathcal{F}^{-1}}{\overset{\mathcal{F}}{\Longleftrightarrow}} e^{-j\,2\pi f t}$$

which gives us

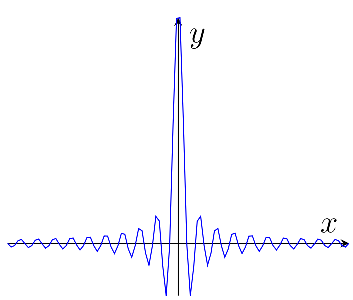
Duality:

$$\begin{aligned}
\mathcal{F}[X(t)] &= \int X(t)e^{-j\,2\pi ft}\,dt \\
&= \int x(u)e^{-j\,2\pi(-tu)}\,du \\
&= \int\int x(u)e^{-j\,2\pi tu}e^{-j\,2\pi t}\,du\,dt \\
&= \int x(u)\left[\int e^{-j\,2\pi t(u+f)}\,dt\right]\,du \\
&= \int x(u)\delta(u+f)\,du \\
&= x(-f)
\end{aligned}$$

Graphically, for a square function,

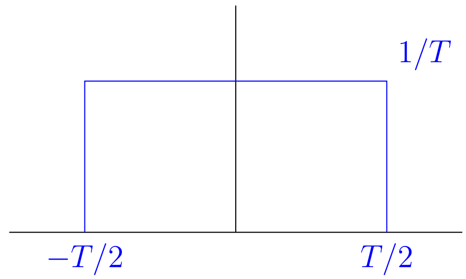


$$\begin{aligned}
\mathcal{F}[x(t)] &= \int_{-T}^{T} e^{-j\,2\pi ft}\,dt \\
&= \left(\frac{1}{-j\,2\pi f}\right)e^{-j\,2\pi ft}\Big|_{-T}^{T} \\
&= -\frac{1}{j\,2\pi f}\left(e^{-j\,2\pi fT} - e^{j\,2\pi fT}\right) \\
&= -\frac{1}{\pi f}\sin(-2\pi fT) \qquad \left(\sin z = \frac{e^{jz}-e^{-jz}}{2j}\right) \\
&= \frac{\sin 2\pi fT}{\pi f}
\end{aligned}$$



1.3 Jan 29

Recall: What is the Fourier transform of this box?



(Note this is an impulse as $T \to 0$)

$$\mathcal{F} = \int_{-T/2}^{T/2} \frac{1}{T} e^{-j\,2\pi f t}\, dt = \frac{1}{T}\left(\frac{1}{-j\,2\pi f}\right) e^{-j\,2\pi f t}\Big|_{-T/2}^{T/2} = \frac{1}{-j\,2\pi f T}\left(e^{-j\,\pi f T} - e^{j\,\pi f T}\right) = \frac{\sin(\pi f T)}{\pi f T}$$

How might we sanity check this?

We already saw that as $T \to 0$, this becomes an impulse in the time domain. What happens in the frequency domain?

$$\lim_{T\to 0} \frac{\sin \pi f T}{\pi f T} = 1$$

and indeed this matches the rule $\mathcal{F}(\delta(t)) = 1$.

Example:

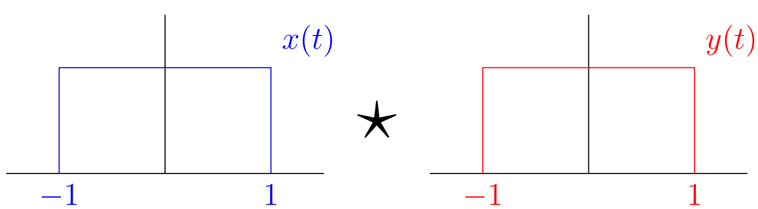
$$\mathcal{F}[e^{-at}u(t)] = \int_0^\infty e^{-at} e^{-j\,2\pi f t}\, dt = \int_0^\infty e^{-(a+j\,2\pi f)t}\, dt = \frac{1}{a + j\,2\pi f}$$

(for $a > 0$ and $u(t)$ the unit step function)

Convolution:

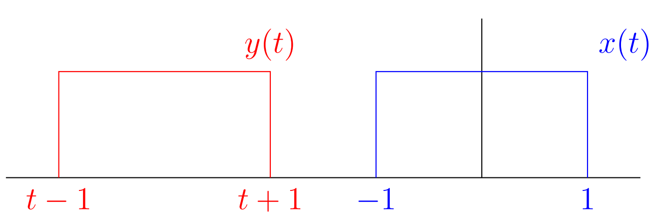
$$\int_{-\infty}^{\infty} x(\tau) y(t-\tau)\, d\tau$$

Example: Convolve two box functions of width 2:

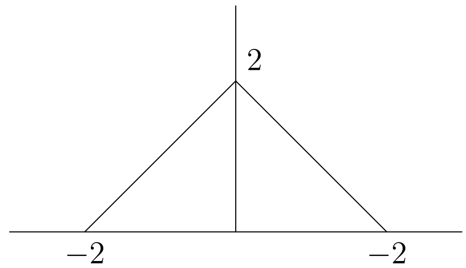


We could do this directly or we can think about it geometrically.

$y(t-\tau)$ corresponds to a reflection and shift by t.

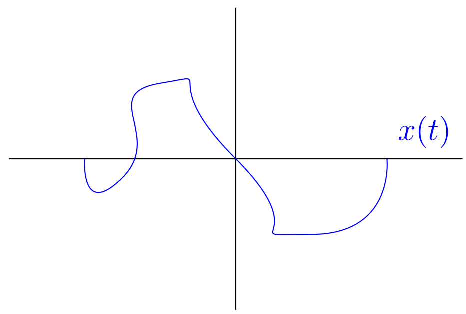


Clearly, multiplying these two functions will give nonzero output only when they overlap. When do they overlap? Precisely at $t \in (-2, 2)$.



Notice that the length of the domain was 2 before convolving and 4 after convolving. This makes sense since convolution is a "spreading and smoothing" operation

Example: $\delta(t) \star x(t)$ where $x(t)$

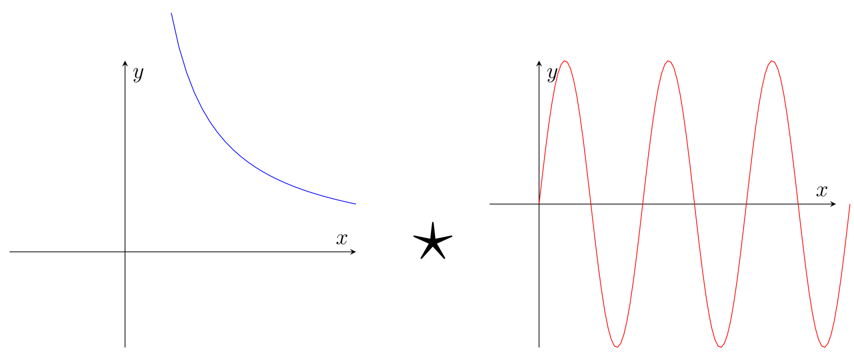


Trick question!

$$\int \delta(x-\tau)x(\tau)\, d\tau = x(t)$$

(sifting!)

For this class, we should think about convolutions critically and graphically:



What is the convolution of these two functions at $t = -2$?

These would be quite annoying to do directly but if we notice that $x(t)$ is undefined for $t \leq 0$ then we can see that there is no overlap at all and the convolution is 0.

Recall:

$$\mathcal{F}(x(t)) = X(f)$$
$$\mathcal{F}(X(t)) = x(-f)$$
$$\mathcal{F}(\delta(t)) = 1$$

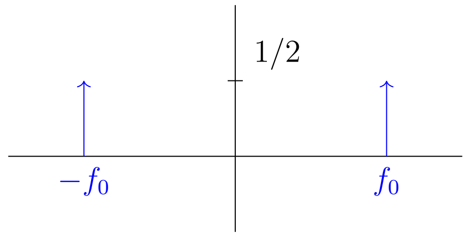
so

$$\mathcal{F}(\delta(t-t_0)) = e^{-j\,2\pi f t_0}$$
$$\mathcal{F}(e^{-j\,2\pi f_0 t}) = \delta(f+f_0)$$

Example: $\mathcal{F}(\cos 2\pi f_0 t)$ does not converge analytically BUT

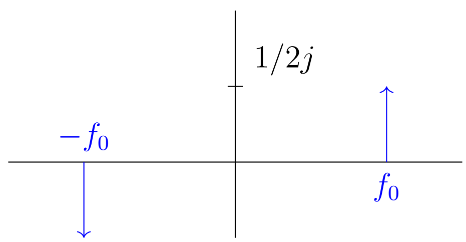
$$\cos 2\pi f_0 t = \frac{e^{j\,2\pi f_0 t} + e^{-j\,2\pi f_0 t}}{2}$$

so from the above,



Similarly,

$$\begin{aligned}\mathcal{F}(\sin 2\pi f_0 t) &= \mathcal{F}\left(\frac{1}{2j} e^{j\,2\pi f_0 t} - e^{-j\,2\pi f_0 t}\right) \\ &= \frac{1}{2j}(\delta(f+f_0) - \delta(f-f_0))\end{aligned}$$



Complex conjugation: $(a+bj)^* = a - bj$

We know

$$\mathcal{F}(x(t)) = \int x(t) e^{-j\,2\pi f t}\,dt = X(f)$$

so what is $\mathcal{F}(x^*(t))$?

$$\begin{aligned}F(x^*(t)) &= \int x^*(t) e^{j\,2\pi f t}\,dt \\ &= \int x^*(t) e^{-j\,2\pi(-f)t}\,dt \\ &= X(-f) \\ &= X^*(f)\end{aligned}$$

leading to the property:

$$\boxed{X^*(f) = X(-f)}$$

Even/Odd Property: Let $x(t) = x(-t)$. Then

$$\int_{-\infty}^{\infty} x(t) e^{-j\,2\pi f t}\,dt = \int_0^{\infty} x(t) e^{-j\,2\pi f t} + x(t) e^{j\,2\pi f t}\,dt = 2\int_0^{\infty} x(t)\cos 2\pi f t\,dt \in \mathbb{R}$$

Even functions have real fourier transforms. Odd functions have imaginary fourier transforms.

1.4 Linear Systems Assessment

Exercise: The Fourier transform of $x(t)$ is $X(f)$. Prove that the Fourier transform of $x(t - t_0)$ is $e^{-j\,2\pi f t_0} X(f)$.

Exercise: The energy in a signal $x(t)$ is defined as

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \; dt$$

Prove that if $X(f)$ is the Fourier transform of $x(t)$, then we must also have

$$E = \int_{-\infty}^{\infty} |X(f)|^2 \; df$$

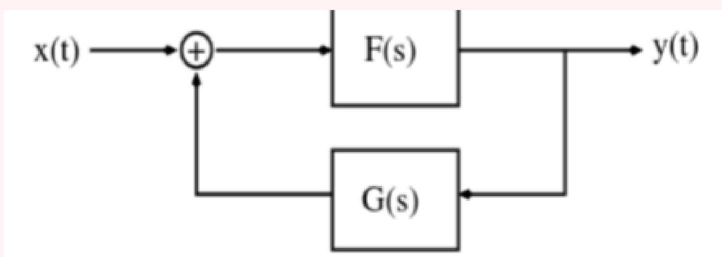
Hint: You may use the fact that

$$\int_{-\infty}^{\infty} e^{j\,2\pi v t} \, dt = \delta(v)$$

Exercise: A signal $x(t)$ of finite energy is applied to a square-law device whose output is defined by $y(t) = x^2(t)$. The spectrum $X(f)$ of $x(t)$ is limited to the frequency interval $-W \le f \le W$. Show that the spectrum, $Y(f)$ of $y(t)$ is limited to the frequency interval $-2W \le f \le 2W$.

Exercise: Show that the overall transfer function $H(s)$ for the feedback system below is given by

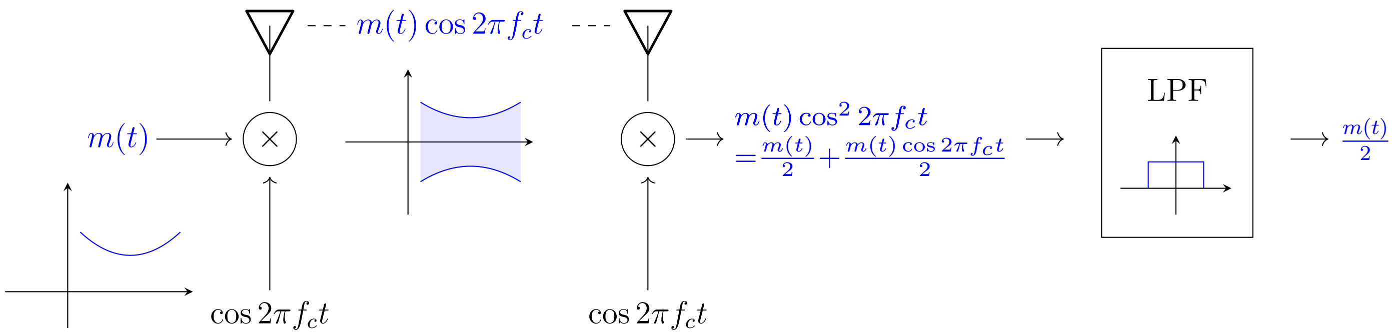
$$H(s) = \frac{F(s)}{1 - F(s)G(s)}$$



Chapter 2

Modulation

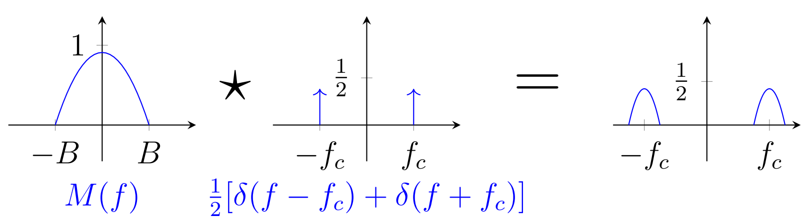
Consider a simple modulation and demodulation scheme:



for $m(t)$ a message and f_c a carrier frequency ($f_c \gg B$, the bandwidth of $m(t)$) which transmits the message better since it is higher frequency.

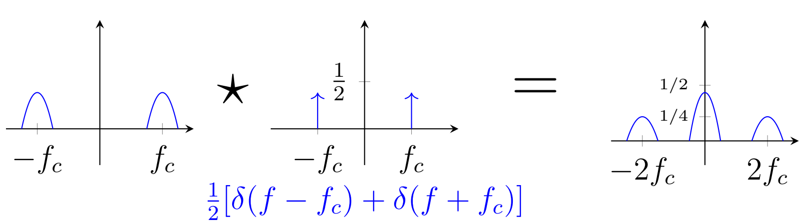
What exactly is happening here? Let's look in the frequency domain:

1. Modulation:



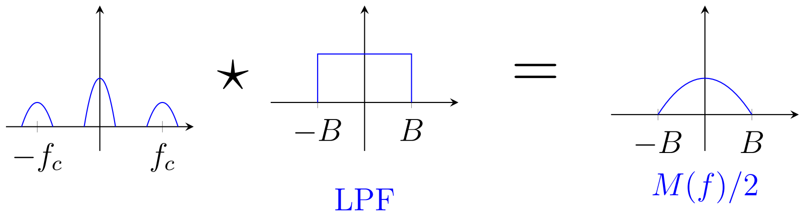
   Convolving with an impulse shifts the signal to be centered at the impulse!

2. Demodulation:



   (spread a copy of the signal around $-f_c$ and f_c so that the interference terms reinforce each other at the baseband)

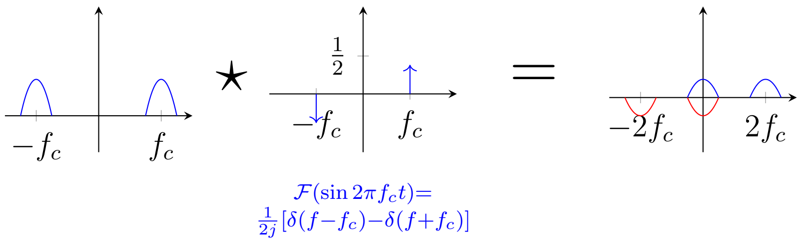
3. Filtering:



2.1 Feb 3

Coherent AM: Synchronous modulation; Modulate with $\cos 2\pi f_c t$ and demodulate with the same signal.

Recall in our demodulation step we multiplied by $\cos 2\pi f_c t$ again. What if we made a mistake and used $\sin(2\pi f_c t)$ instead?

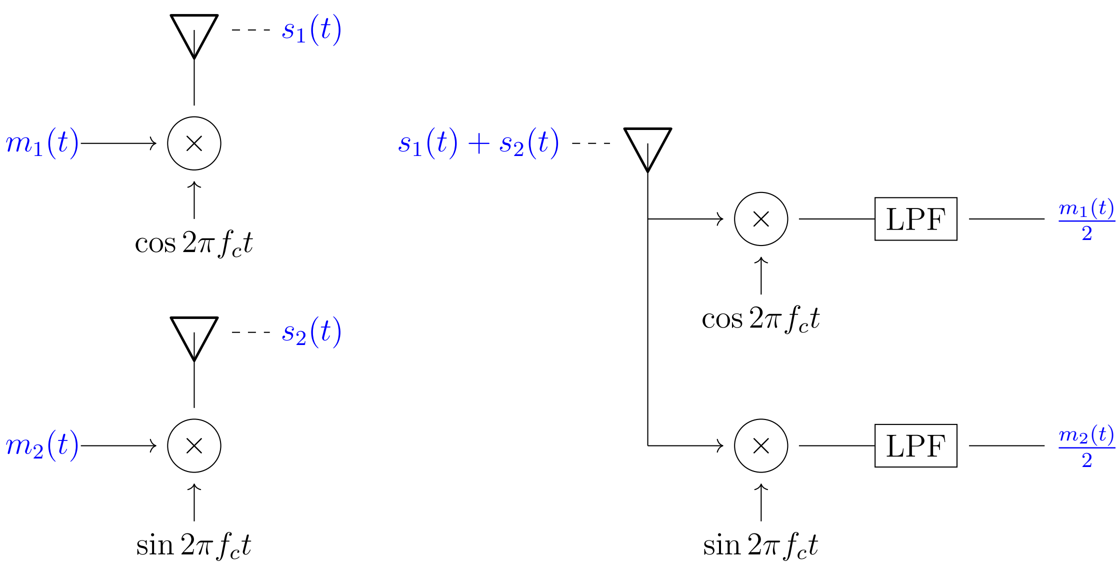


So the signals cancel around the baseband and we get no message after filtering!

How could we recover if we made this mistake?

Let's try broadcasting two messages $m_1(t)\cos 2\pi f_c t$ and $m_2(t)\sin 2\pi f_c t$ simultaneously.

The received signal is $(s_1(t) + s_2(t))$. If we demodulate by $\cos 2\pi f_c t$ on one rail and $\sin 2\pi f_c t$ on the other rail and filter, we get two separate messages back, $m_1(t)/2$ and $m_2(t)/2$ respectively!



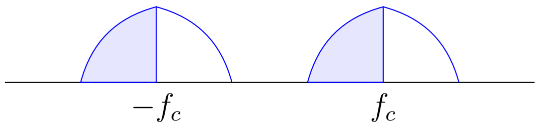
Single sideband modulation: Let $m(t)$ be a real signal with Fourier transform $M(f)$. We know

$$M(f) = \int m(t)e^{-j\,2\pi f t}\,dt$$
$$M^*(f) = \int m(t)e^{-j\,2\pi(-f)t}\,dt = M(-f)$$

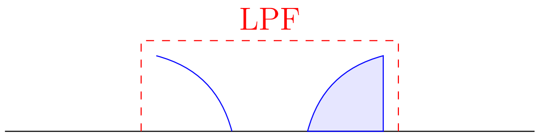
(i.e. the real part is even and the imaginary part is odd)

In particular, this means we can send less information by sending only the positive frequencies or only the negative frequencies and reconstructing the other half using the even/odd properties.

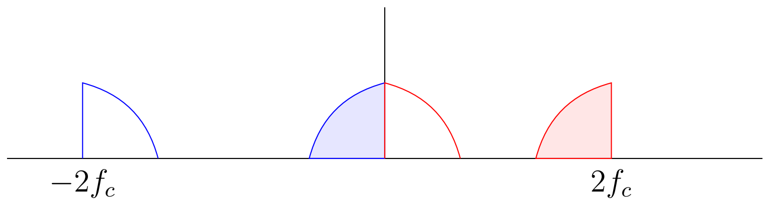
Consider:



We filter,

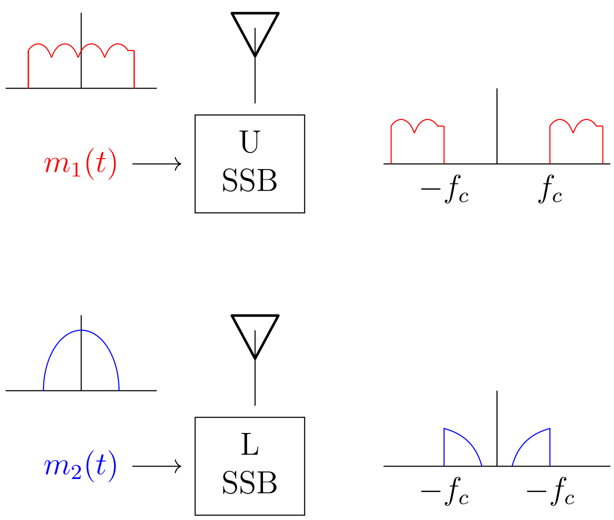


Then modulate by $\cos 2\pi f_c t$:



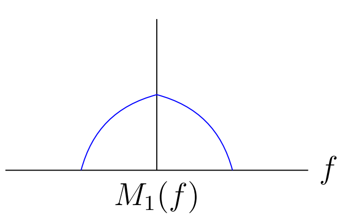
and just like that, we have recovered the full signal using half the bandwidth!

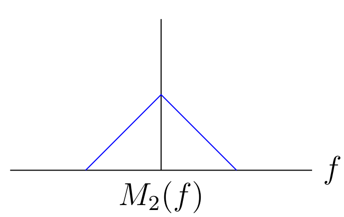
In practice, we can use two signals (an upper and lower sideband):

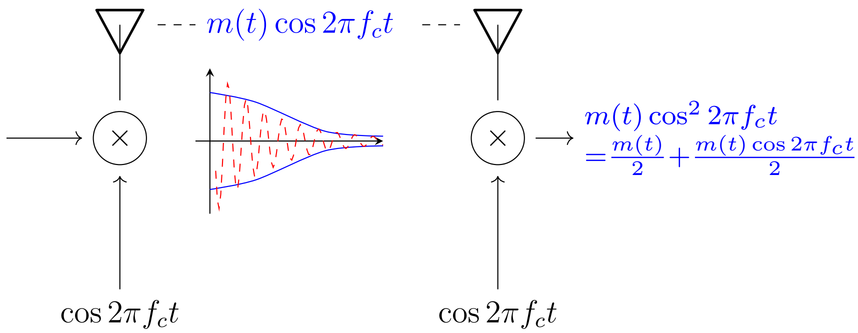


Frequency Division Multiplexing:

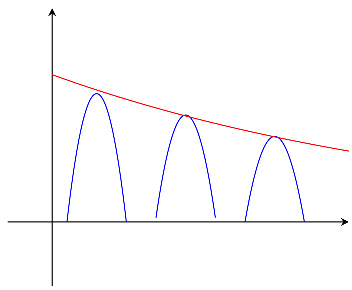
Define two signals $m_1(t)$ and $m_2(t)$.



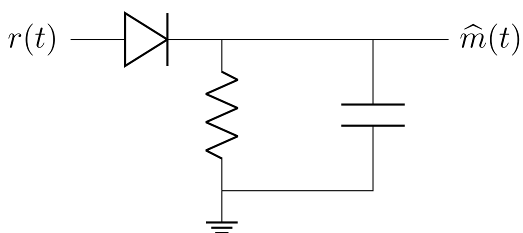




Though we could also blow up in time and rectify the transmitted wave



Though we still want to recover the envelope (all the information is in the amplitude of the envelope).



Motivating FM:

$$r(t) = \sin\left(2\pi f_c t + \int_{-\infty}^{t} m(\tau) d\tau\right)$$

Where is the information here? In the integral.

What does it look like? First, we know it has constant amplitude ("modulus 1").

Define the *instantaneous frequency*

$$\omega(t) = \frac{d}{dt}\Theta(t) = 2\pi f_0 = \omega_0$$

for $\Theta(t) = 2\pi f_0 t$ the *instantaneous phase* of the signal.

Now let's try differentiating:

$$\frac{d}{dt} r(t) = (2\pi f_c + m(t)) \cos\left(2\pi f_c t + \int_{-\infty}^{t} m(\tau)\, d\tau\right)$$